INTRODUCTION TO COMPUTER GRAPHICS

What is Computer Graphics?

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What is Computer?

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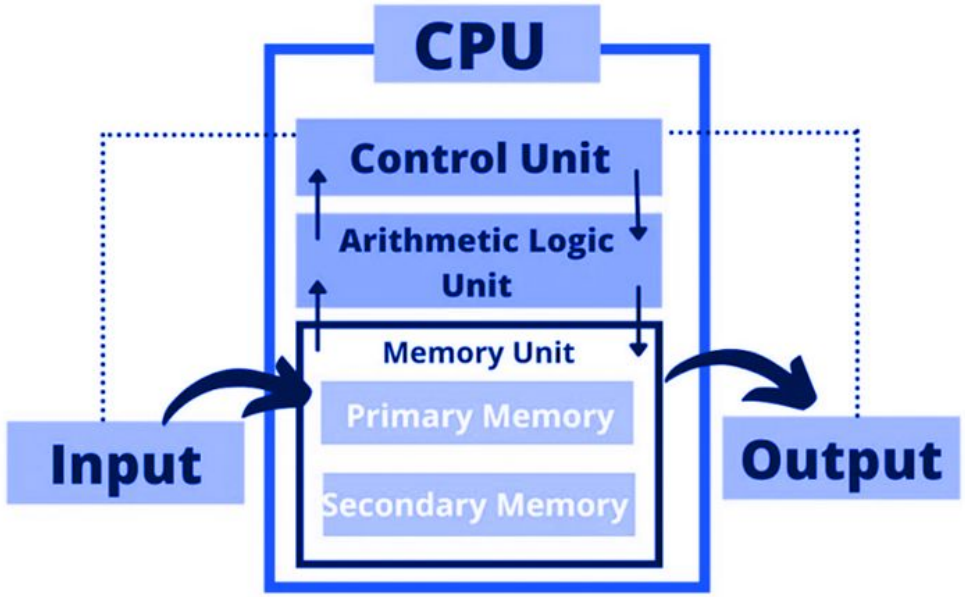

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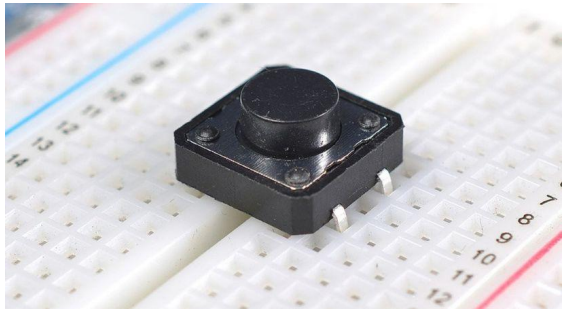

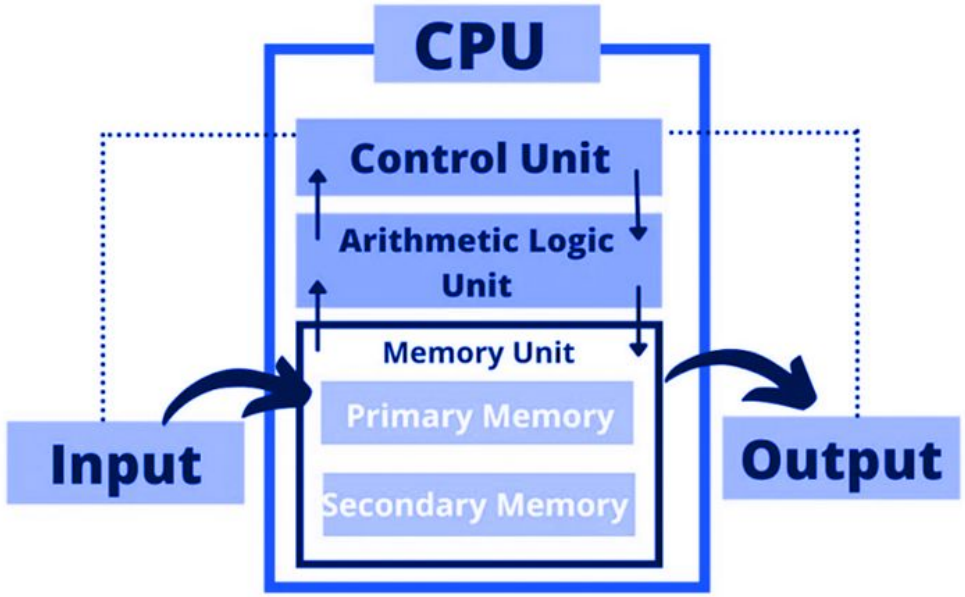

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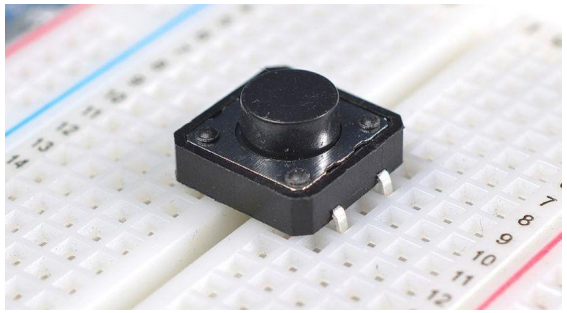

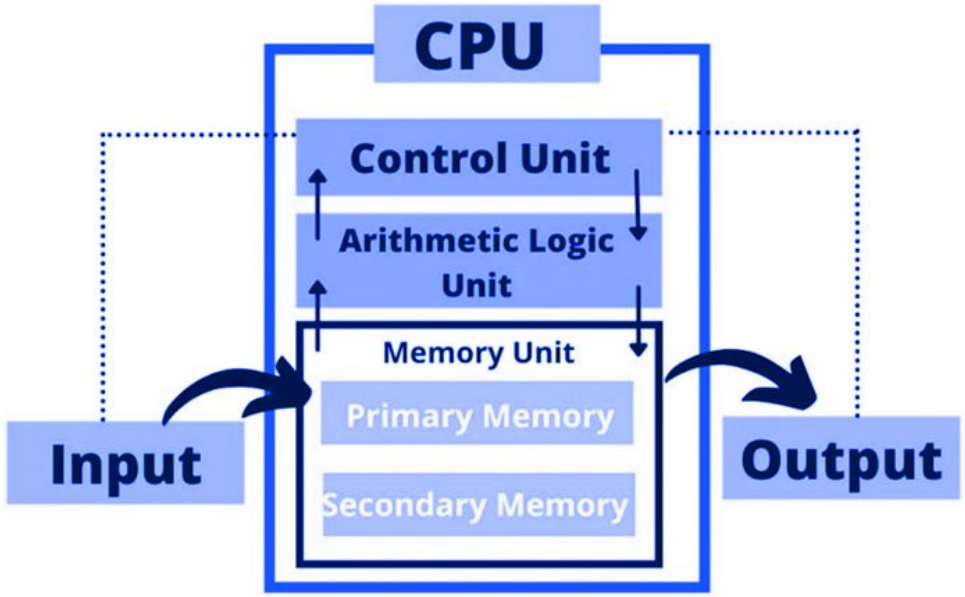

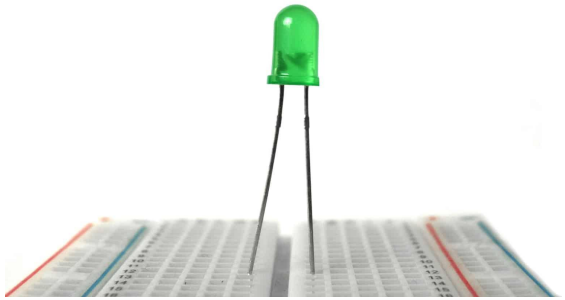

What is Computer Graphics?

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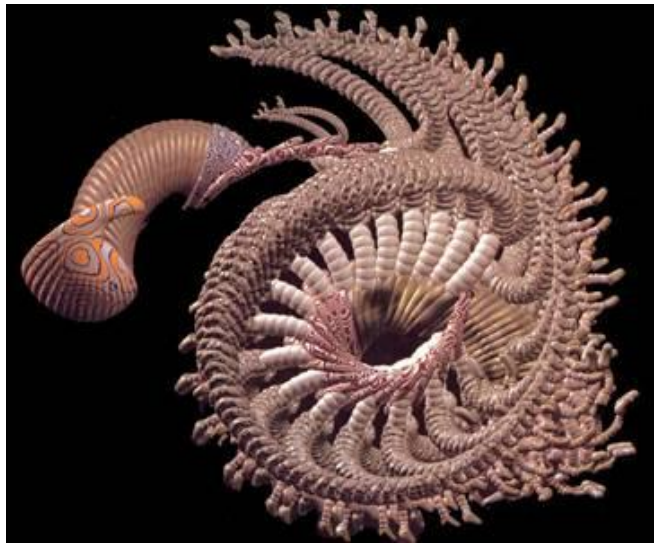
What is Computer Graphics?

Creating visuals with the help of a computer

What is Computer Graphics?

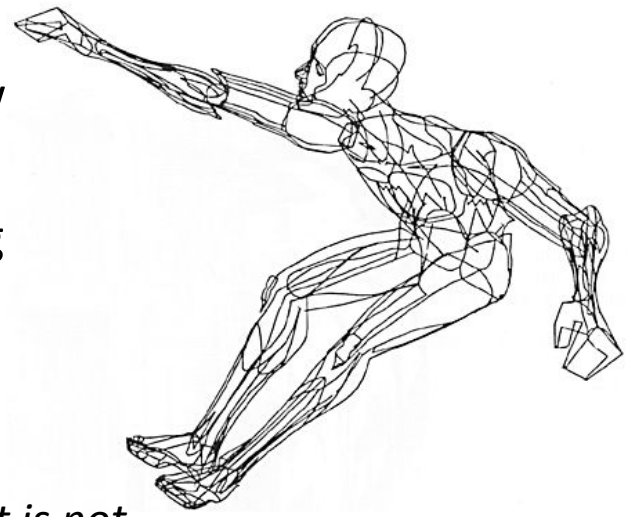
- Computer graphics generally means creation, storage and manipulation of models and images
- Such models come from diverse and expanding set of fields including physical, biological, mathematical, artistic, and conceptual/abstract structures

Frame from animation by William Latham, shown at **SIGGRAPH 1992**. Latham creates his artwork using rules that govern patterns of natural forms.



What is Computer Graphics?

- William Fetter coined term “computer graphics” in 1960 to describe new design methods he was pursuing at Boeing for cockpit ergonomics
- Created a series of widely reproduced images on “pen plotter” exploring cockpit design, using 3D model of human body.

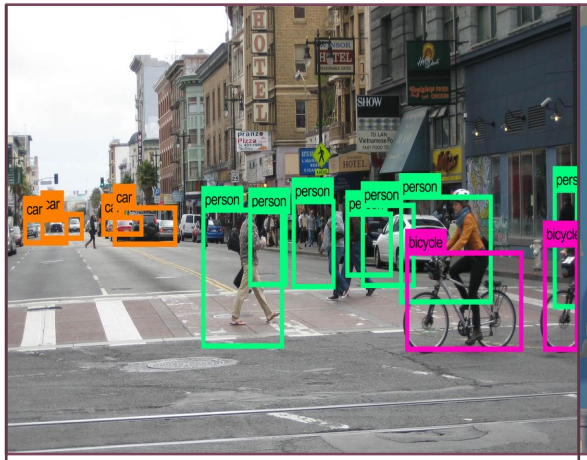
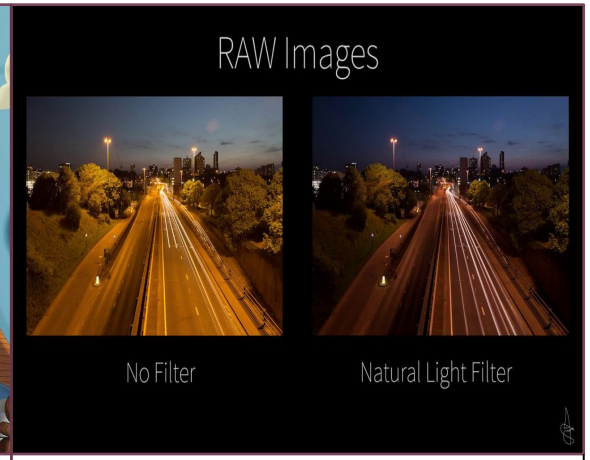


“Perhaps the best way to define computer graphics is to find out what it is not. It is not a machine. It is not a computer, nor a group of computer programs. It is not the know-how of a graphic designer, a programmer, a writer, a motion picture specialist, or a reproduction specialist.

Computer graphics is all these – a consciously managed and documented technology directed toward communicating information accurately and descriptively.”

Computer Graphics, by William A. Fetter, 1966

Differences

		
Computer vision: **Extracting information** from the contents of an input image or video frame. Ex: Face Recognition, Autonomous Driving	Computer graphics: **Creating** an image **from scratch** using computer. Ex: Animated Movies	Digital image processing: **Processing raw input images** to perform different operations. Ex: Apply Filter on an Image, Noise Reduction, Compression

Applications of CG

1. **Entertainment & Gaming:**

   Visual Effects, Animation, Video Games, VR/AR Experiences.

2. **Design & Engineering:**

   CAD Modeling, Architectural Visualization, Product prototyping.

3. **Education & Training:**

   Virtual Labs, Interactive simulations, VR-based Learning.

4. **Medical & Scientific Visualization:**

   MRI/CT Imaging, Surgery Simulation, Data and Astrophysical Visualization.

5. **User Interfaces & Marketing:**

   GUIs, Graphic Design, 3D Product Visualization, Digital Ads.

Input Methods in Computer Graphics

- **Keyboard and Mouse**: Traditional devices for controlling software, navigation, and object manipulation.
- **Graphics Tablets**: Used for digital drawing and illustration.
- **Touchscreens**: For direct interaction with graphical elements.
- **3D Scanners**: Capture the 3D structure of objects to create digital models.
- **Body as Interaction Device**:

Xbox Kinect

Leap Motion

Nimble UX

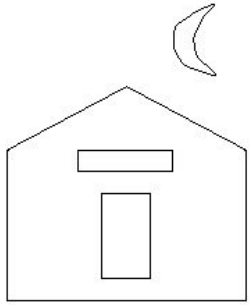
Output Devices in Computer Graphics

- **Monitors and Screens**: The most common output devices, used to display images and videos.
- **Printers**: For generating hard copies of graphics (2D or 3D).
- **Projectors**: Used in large-scale displays for presentations and entertainment.
- **VR/AR Headsets**: Immersive displays for virtual and augmented reality applications.

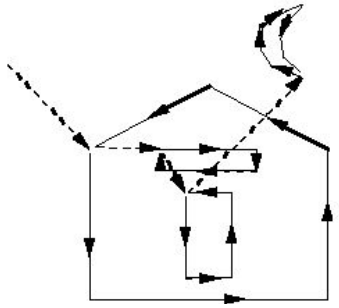
Graphics Display Hardware

Vector Display (calligraphic, stroke, random-scan)

- Driven by display commands
  - (move (x, y), char(“A”) , line(x, y)…)
- Survives as “scalable vector graphics”



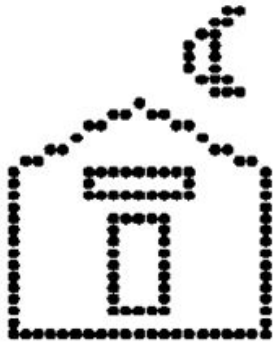
Ideal Drawing



Vector Drawing

Raster Display (TV, bitmap, pixmap) used in displays and laser printers

- Driven by array of pixels (no semantics, lowest form of representation)
- Note “jaggies” (aliasing errors) due to discrete sampling of continuous primitives



Outline

Filled

Output Technology: Vector Displays

- Also called Calligraphic, Stroke or Line Drawing Graphics
- Lines drawn directly on phosphor
  - Display processor directs electron beam according to list of lines defined in a "**display list**"
  - Phosphors glow for only a few microseconds so lines must be redrawn or **refreshed** constantly
  - Deflection speed limits # of lines that can be drawn without flicker.

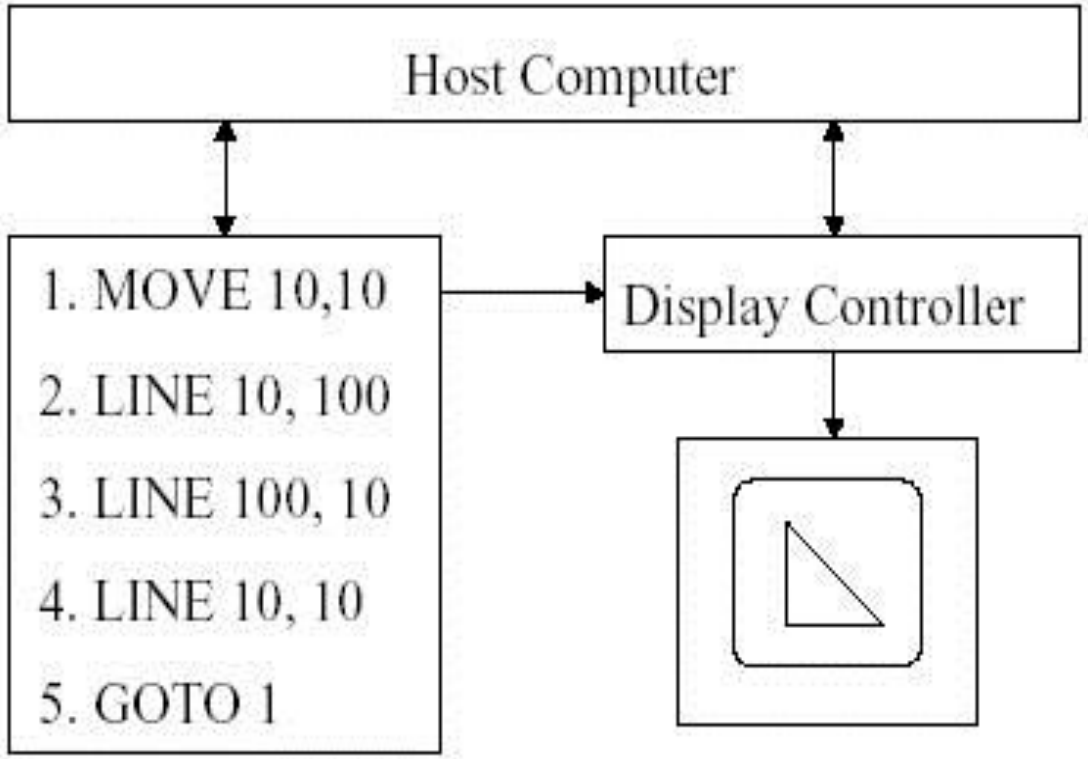


Fig.: Vector Display Architecture

Output Technology: Raster Displays

- Display **primitives** (lines, shaded regions, characters) stored as pixels in **refresh buffer** (or **frame buffer**)
- Electron beam scans a regular pattern of horizontal raster lines connected by horizontal retrace and vertical retrace
- Video controller coordinates the repeated scanning
- Pixels are individual dots on a raster line

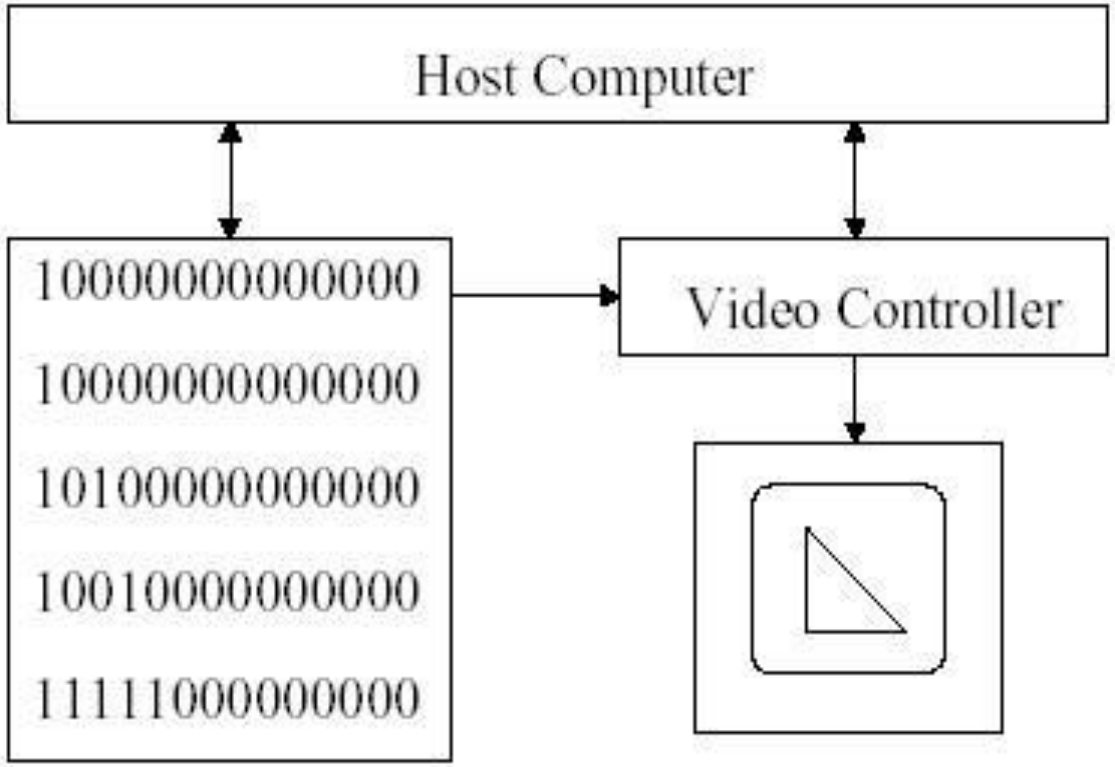
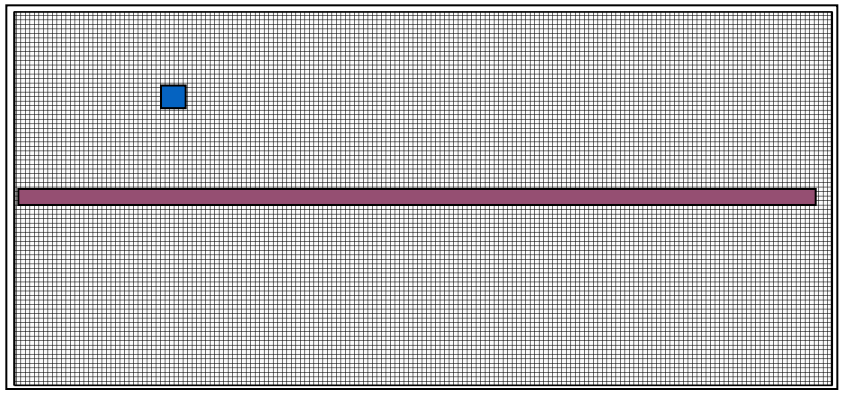


Fig.: Raster Display Architecture

Basic Definitions

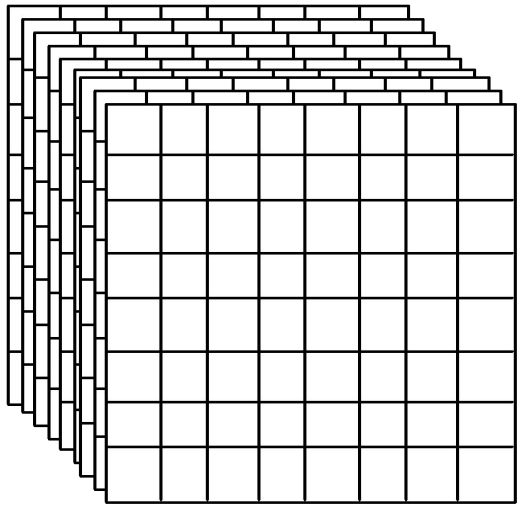
- **Raster:** A rectangular array of points or dots.
- **Pixel (Pel):** One dot or picture element of the raster
- **Pixel Grid:** Images are arranged in a grid of pixels, where each pixel is stored with specific color information (RGB values)
- **Scan line:** A row of pixels



Video raster devices display an image by sequentially drawing out the pixels of the scan lines that form the raster.

Frame Buffer

- A frame buffer is characterized by its size, x, y, and pixel depth.
- the **resolution** of a frame buffer is the number of pixels in the display. e.g. 1024x1024 pixels.
- Bit Planes or Bit Depth is the number of bits corresponding to each pixel. This determines the **color resolution** of the buffer.
- Dual ported (simultaneously writing values and displaying in the monitor)



Bilevel or monochrome displays have 1 bit/pixel (128Kbytes of RAM)
8bits/pixel -> 256 simultaneous colors
24bits/pixel -> 16 million simultaneous colors

Raster vs. Vector Displays

- **Raster**: Pixel-based, fixed resolution, common in displays. Cheaper. More realistic.
- **Vector**: Shape-based, resolution-independent, best for scalable designs.
- **Use Cases**: Raster is used for photographs and complex images, while vector is used for fonts and logos.

Rendering Pipeline

The rendering pipeline is a sequence of steps used in computer graphics to convert a 3D scene into a 2D image on the screen.

Steps of the Rendering Pipeline:

1. **Vertex Processing & Clipping**
   - Transforms *object coordinates → world → camera → screen coordinates* using **model, view, and projection transformations**.
   - Performs **lighting calculations** (*per-vertex lighting* in fixed pipelines).
   - **Clipping:** Removes objects or parts of objects outside the camera can see (**view frustum**)

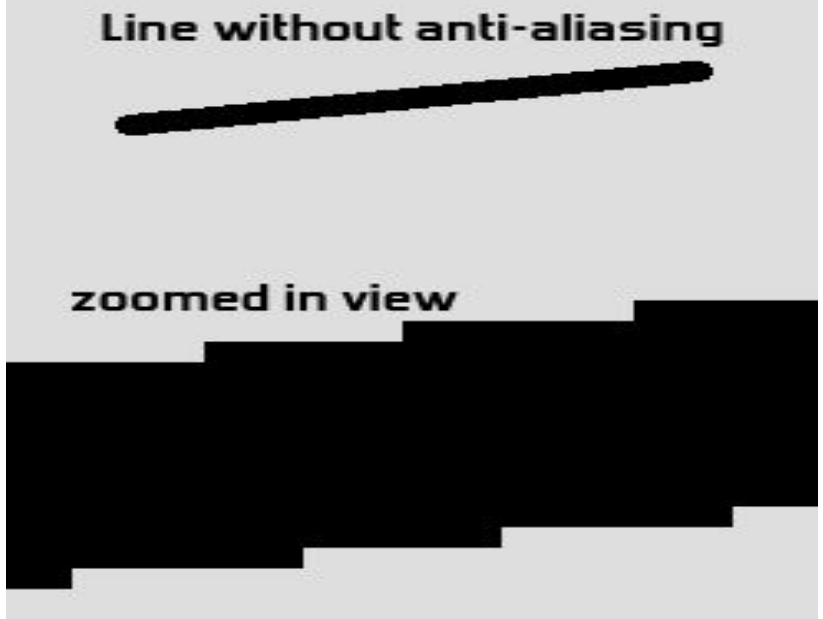
1. **Rasterization**
   - Converts 3D primitives (like triangles, lines) into **2D fragments (potential pixels)**.
   - Determines which pixels are covered by each primitive and passes them to the next stage.

Rendering Pipeline

3. **Fragment (Pixel) Processing**
   - Applies **shading, textures, and lighting** (*per-pixel lighting* in modern shader pipelines).
   - Performs **depth testing, transparency, reflection, and other pixel-level effects**.
4. **Image Composition / Output Merging**
   - Combines fragments from all objects into the **final image**.
   - Applies **depth buffering (Z-buffer)**, **blending**, and **anti-aliasing**.
   - Outputs the result to the **framebuffer** for display on the screen.

Antialiasing

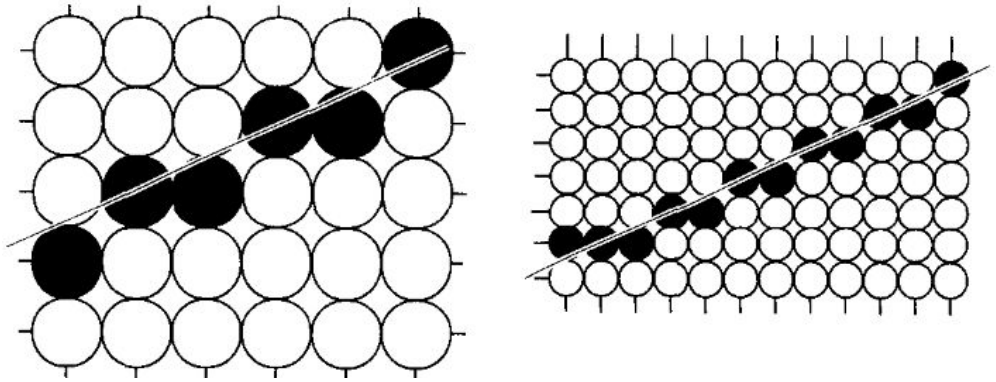
- Aliasing
  - Approximating a continuous entity with discrete samples
  - Jagging / staircasing effect
  - Result of an all-or-nothing approach to scan conversion
    - Each pixel is either colored or left unchanged



- Antialiasing
  - The application of techniques that reduce or eliminate aliasing

Antialiasing Techniques

- Increase Screen Resolution
  - Costly
    - Only diminishes, does not solve
    - Higher memory
    - More time for scan-conversion



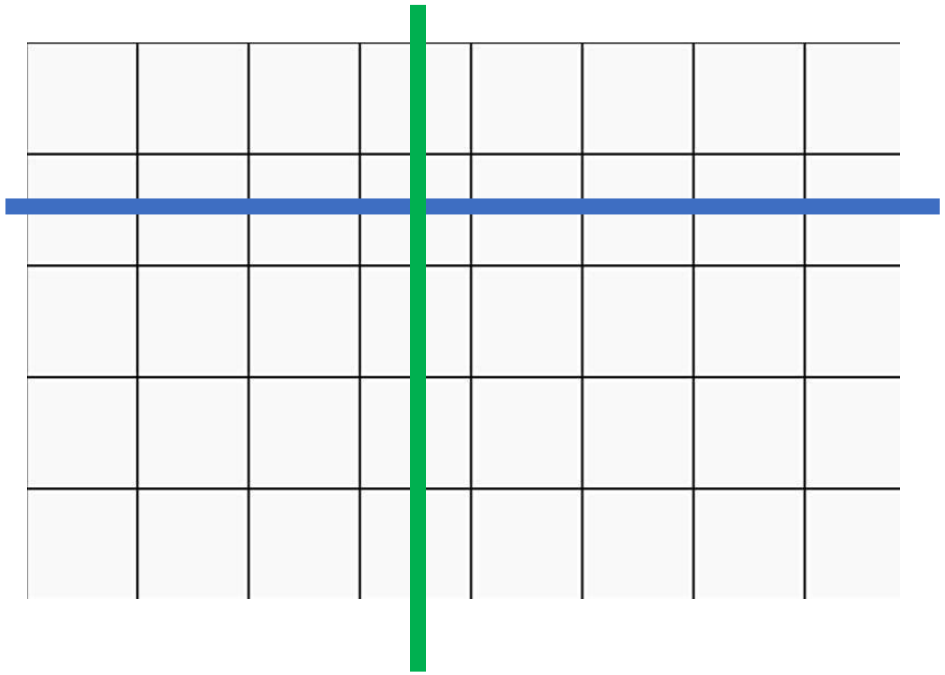
(a) res. W x H (b) res. 2W X 2H

- Area Sampling
  - Unweighted Area Sampling
  - Weighted Area Sampling

Unweighted Area Sampling

Basic Idea:

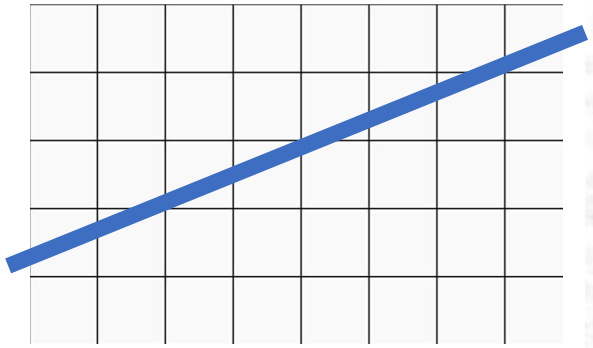
- Horizontal or vertical line passes through one pixel per row or column

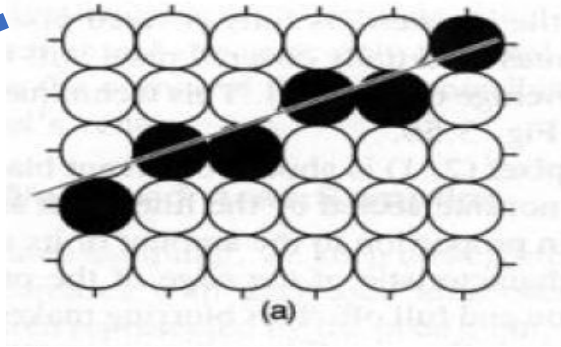


Unweighted Area Sampling

Basic Idea:

- Horizontal or vertical line passes through one pixel per row or column
- **Lines at other angles go through multiple pixels per row or column**





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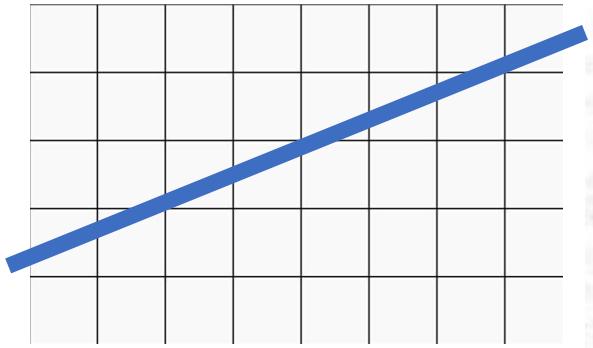

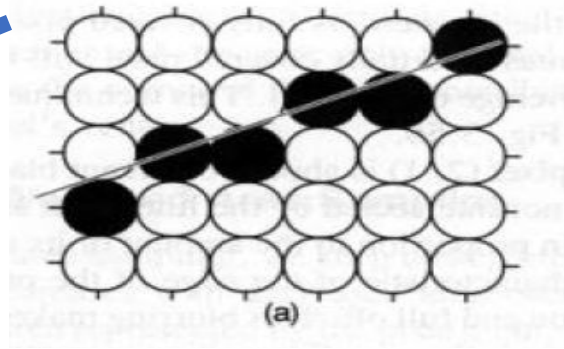

- **A line is considered as a rectangle of a desired thickness covering a portion of the grid**

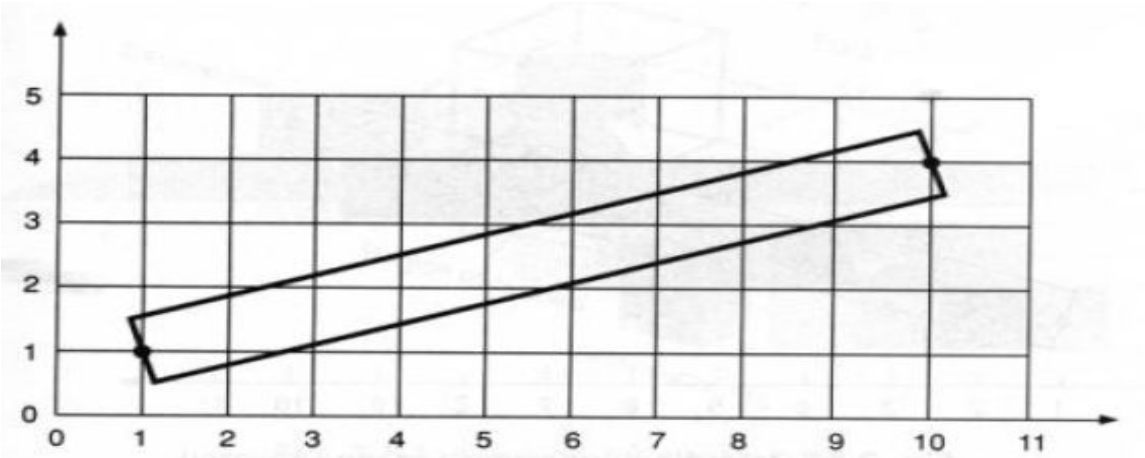

Figure 3.55 Line of nonzero width from point (1,1) to point (10,4).

Unweighted Area Sampling

Basic Idea:

- Pixels on the grid overlapped by the rectangle are colored to an appropriate intensity. General rule : A line contributes to each pixel's intensity an *amount proportional to the percentage* of the pixel's tile it covers.

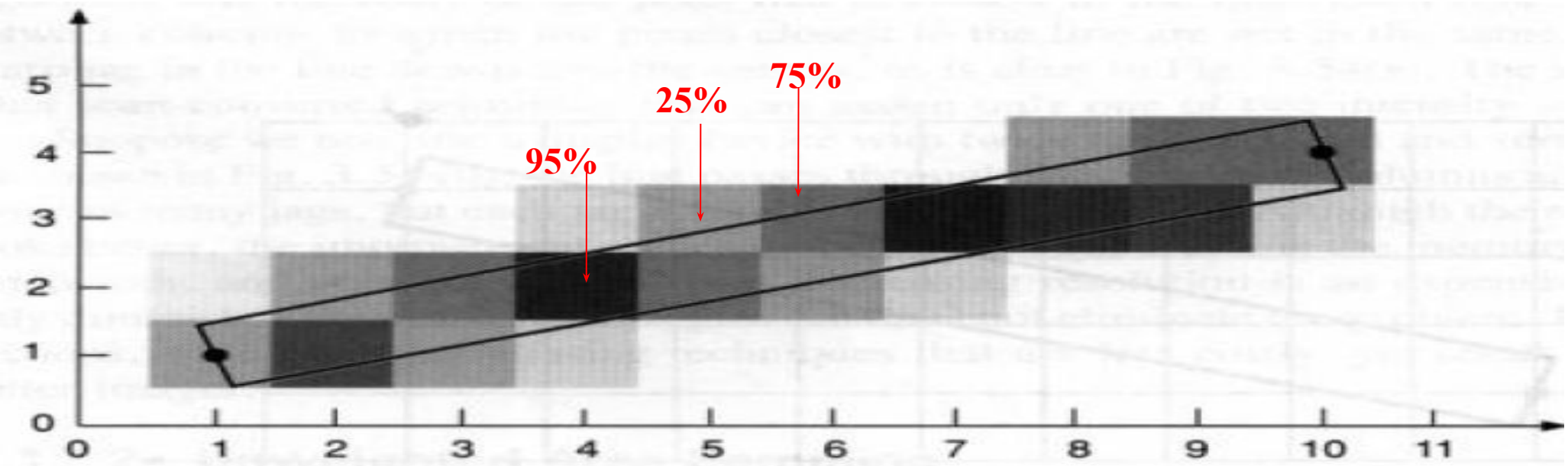

Fig. 3.56 Intensity proportional to area covered.

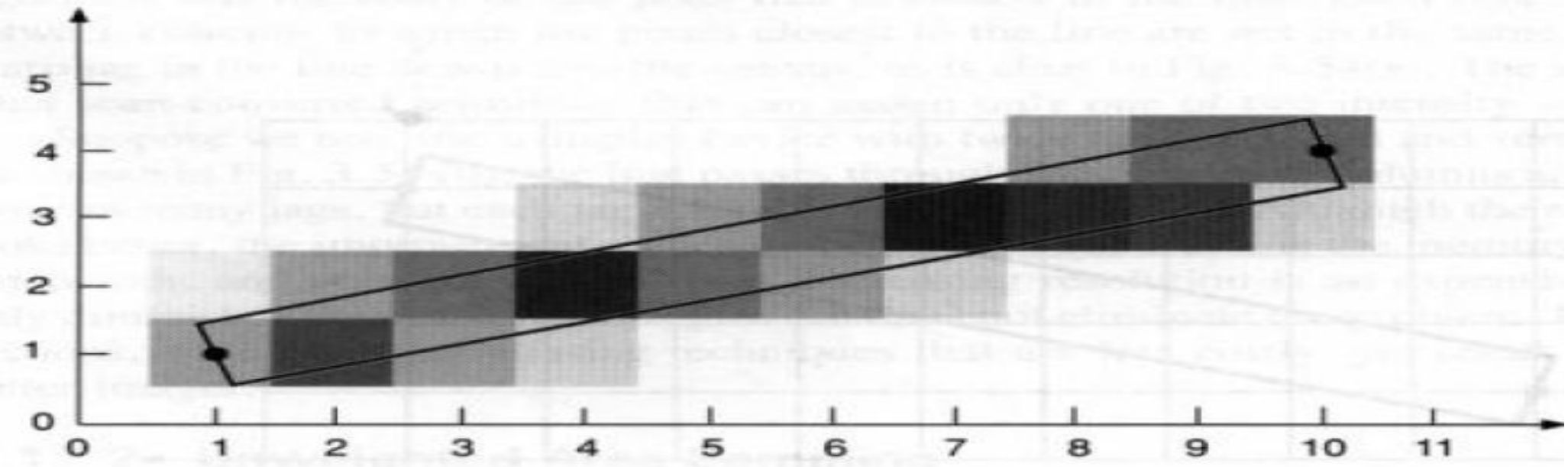

Fig. 3.56 Intensity proportional to area covered.

Intensity of pixel centered at (x,y) :

$I_{x,y} = I_{max} \cdot dA \cdot \text{Weight}$

dA = area overlap for pixel at (x,y)

Weight = 1 for unweighted area sampling